

JONATHAN KHO

Department of Astronomy, University of Virginia $\diamond$ yja6qa@virginia.edu

EDUCATION

University of Virginia

2024 - Present

Doctorate of Philosophy in Astronomy

Master's of Science in Astronomy (2026)

Michigan State University

2020 - 2023

Bachelor's of Science in Astrophysics

Minor in Data Science

Minor in Computational Mathematics, Science, and Engineering

SELECT PUBLICATIONS

First Author

- **Kho**, Bhowmick, Weinberger, Torrey, Blecha, Hernquist, Bryan, Garcia, Ahvazi, Saravia, Oh (2026) [*Learning the Universe at High Redshifts*], Submitted.
- **Kho**, Bhowmick, Torrey, Garcia, Ahvazi, Blecha, Vogelsberger (2025) ["Signatures of Black Hole Seeding on the $M_{\bullet}$ - σ Relation: Predictions from the BRAHMA Simulations"] ApJ 94.2, 172, p. 172.

Co-Author

- Garcia, Rose, Torrey, Caputo, Lisanti, Pace, Liu, Hussein, Liu, Villaescusa-Navarro, Barry, Leisher, Costanza, **Kho**, Lilie, Li, Ahvazi, Bhowmick, Nguyen, O'Neil, Ou, Shen, Farahi, Kallivayalil, Necib, Vogelsberger (2025) ["The DREAMS Project: Disentangling the Impact of Halo-to-Halo Variance and Baryonic Feedback on Milky Way Dark Matter Density Profiles"] ApJ. Submitted.
- Ahvazi, Pace, Garling, Ou, Kallivayalil, Torrey, Benson, Bhowmick, Torres-Albà, Garcia, Saravia, **Kho**, Warfield, Atzberger (2025) ["The abundance and properties of the lowest luminosity dwarf galaxies around the Milky Way: Insights from Semi-Analytic Models"] ApJ. Submitted.
- Bhowmick, Blecha, Torrey, Kelley, Natarajan, Somerville, Weinberger, Garcia, Hernquist, Di Matteo, **Kho**, Vogelsberger (2025) ["Heavy seeds and the first black holes: Insights from the BRAHMA simulations"] ApJ. Submitted.
- Bhowmick, Blecha, Torrey, Somerville, Kelley, Weinberger, Vogelsberger, Hernquist, Natarajan, **Kho**, Di Matteo (2025) ["Signatures of black hole seeding in the local Universe: predictions from the BRAHMA cosmological simulations"] MNRAS 538.1, pp. 518–536.

CONTRIBUTED TALKS AND PRESENTATIONS

- Cosmology & Galaxy Astrophysics with Simulations and Machine Learning 2027 - New York, NY
January 2027
- Building Galaxies from Scratch II - Garching, Germany
September 2026
- Cosmic Horizons Conference [Poster] - Charlottesville, VA, US
July 2026
- UVA Research Computing Exhibition [Poster] - Charlottesville, VA, US
April 2026
- 243rd AAS Meeting [Poster] - New Orleans, LA, US
January 2024

RESEARCH EXPERIENCE

- DREAMS of Galxies on FIRE
2025 - Present

- Building a suite of halos with varying target halo masses and cosmological, astrophysical, and numerical parameters using the FIRE-2 galaxy formation model.
- Black Hole Growth at High Redshift: Impact of Feedback, Accretion, and Mergers 2025 - 2026
 - Investigating the growth of black holes at high redshift with a variety of accretion models, galaxy environments, and feedback strengths in hydrodynamic galaxy simulations. Submitted
- Signatures of black hole seeding on the $M_{\bullet} - \sigma$ relation 2024 - 2025
 - Studied the impact of different black hole seeding criterion on the $M_{\bullet} - \sigma$ relation and its redshift evolution. Published.
- 3D Moving-Mesh Simulations of Type IIIn Supernovae 2023 - 2024
 - Modeled the evolution of type IIIn supernovae with different circumstellar material configurations using the moving-mesh code Arepo. This project was part of an NSF REU internship and resulted in a poster presentation at the 243rd AAS meeting.
- Modeling a rare explosion from the neutron star MAXI J0556-332 2022 - 2023
 - Modeled a proposed thermonuclear explosion of oxygen or neon deep in the crust of a neutron star. Corresponding poster presentation was awarded a UURAF First Place Award.
- Optimizing Accelerator Beam Quality using Reinforcement Learning 2022
 - Used reinforcement learning to adjust magnetic quadrupole field strengths in a simulated accelerator beam to refine the beam quality.

TEACHING

University of Virginia

- ASTR 3490 - Galaxies Spring 2026
- Constellation Quiz Spring 2025
- ASTR 1250 - Alien Worlds Spring 2025
- ASTR 3470 - Science and Controversy in Astronomy Fall 2024
- ASTR 1210 - Introduction to the Sky and the Solar System Fall 2024

Michigan State University

- CMSE 402 - Data Visualization Spring 2024
- CMSE 381 - Fundamentals of Data Science Methods Spring 2024
- PHY 232C - Introductory Physics II Fall 2022, Spring 2023

ACADEMIC AWARDS AND ACHIEVEMENTS

- Explore ACCESS program 2026
 - Awarded an NSF Explore ACCESS program consisting of 400,000 credits to be used for research compute at partnering HPC systems.
- Virginia Space Grant Consortium Fellow 2026
 - Fellowship designed to support individuals pursuing careers in STEM which support NASA's mission with a \$6000 stipend ([link here](#))
- UVA Research Computing Poster Exhibition, 2nd place 2026
 - Won 2nd place in a poster research computing exhibition, earning a \$2000 travel award.

- Board of Trustees Award for Academic Excellence 2023
 - Michigan State award to graduates who earned a perfect GPA average for all of their coursework.
- UURAF First Place Award 2023
 - First Place Award for poster presentation of neutron star crust research at Michigan State's University Undergraduate Research and Arts Forum.
- Lawrence W. Hantel Endowed 2021-2023
 - Michigan State Dept. of Physics and Astronomy award recognizing outstanding undergraduates doing research in physics or astronomy.

SERVICE

- UVA Astronomy Graduate-Undergraduate Mentorship Program 2024 - Present
 - Began and lead a Graduate-Undergraduate Mentorship program enabling Astronomy undergraduates at UVA to receive one-on-one mentorship from an Astronomy graduate student.
- Dark Skies Bright Kids 2024 - 2025
 - Member of the Astronomy outreach and education group.
- Michigan State Astronomy Undergraduate Mentorship Program 2023 - 2024
 - Served as a mentor for more junior undergraduates studying Astrophysics at MSU.

STUDENT MENTORING

C. Surovell	2026 - 2026
L. Compton	2025 - 2026
M. Kulkarni	2025 - 2026
H. Nguyen	2023 - 2024

SKILLS

Python, Git, $\text{\LaTeX}$, HPC, Bash/Slurm